

NON-EXISTENCE OF COLLATZ m -CYCLES FOR $m \leq 95$

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Let C be the accelerated Collatz map, and let an m -cycle denote a non-trivial periodic orbit with m local minima. We prove, assuming the stated certified external inputs, that no non-trivial Collatz m -cycle exists for $m \leq 95$. The proof combines the verification bound $X = 2^{71}$, the Simons–de Weger bounds for the total number of odd steps, a suffix-balanced cyclic block estimate, continued-fraction denominator tests for $\log_2 3$, and exact integer finite searches for the remaining cases $m = 92, 93, 94, 95$. The finite searches exclude the required windows for the least local minimum, and the resulting improved lower bounds for all local minima lead to final continued-fraction contradictions. All finite checks use integer arithmetic, and all logarithmic comparisons are certified by rational intervals.

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1. Introduction

We use the accelerated Collatz map

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

The usual cycle $1, 2$ is called trivial. A periodic orbit different from this one will be called non-trivial.

Following the notation used by Simons–de Weger and Hercher, a non-trivial cycle is decomposed into blocks beginning at local minima. The number of local minima in one period is denoted by m , and the total number of odd steps in the period is denoted by K . The problem considered here is to exclude possible m -cycles for small values of m .

The general structure of the argument goes back to the Diophantine estimates of Simons and de Weger [6]. Their work gives a strong upper bound for K in terms of m . On the other hand, lower bounds for the entries of a hypothetical cycle give lower bounds for K , after one passes through rational approximations to $\log_2 3$. Hercher used this strategy to exclude all non-trivial cycles with $m \leq 91$ [2].

The argument has two parts. First, we give a block estimate sufficient to reproduce the lower-bound iteration for $m \leq 91$. The block is chosen by a suffix-balanced cyclic selection lemma. This form is useful because it gives lower bounds for the local minima that occur later in the selected block, rather than only a lower bound for the total odd-run length of the block.

Second, we treat the next four cases. For $m = 92, 93, 94, 95$, the lower-bound iteration alone does not immediately contradict the Simons–de Weger upper bound. We therefore add finite searches for the possible global least local minimum $n_{\min}$. The ranges excluded are

$$X \leq n_{\min} \leq \frac{73}{10}X, \quad X \leq n_{\min} \leq 15X,$$

and, for both $m = 94$ and $m = 95$,

$$X \leq n_{\min} \leq 24X.$$

After these exclusions, the same continued-fraction test gives contradictions in all four cases.

All numerical comparisons used in the proof are made with rational certificates. The congruence searches use only integer arithmetic.

2. Certified Inputs and Notation

Throughout the paper C denotes the accelerated map above. We assume, for contradiction, that a non-trivial periodic orbit exists. Let

$$n_1, n_2, \dots, n_m$$

be the local minima in one period, indexed cyclically. After n_i , let k_i be the exact number of consecutive odd steps before the next local minimum, and let ℓ_i be the exact number of following even steps. We write

$$K = \sum_{i=1}^m k_i, \quad L = \sum_{i=1}^m \ell_i, \quad \delta = \log_2 3.$$

For a local minimum n_i , put

$$T(n_i) = \sum_{t=0}^{k_i-1} \frac{1}{C^t(n_i)}.$$

The proof uses the following certified numerical inputs. They are external inputs and are not reproved in this paper.

(I1) Every element of a non-trivial cycle is larger than

$$X = 2^{71}.$$

This follows from the computational verification of the Collatz conjecture up to this bound [1].

(I2) For every non-trivial cycle,

$$K > 7 \cdot 10^{11}.$$

This lower bound is due to Simons and de Weger [6].

(I3) For every non-trivial m -cycle,

$$K < 1.4784 m \delta^m.$$

This upper bound is also due to Simons and de Weger [6].

Inputs (I2) and (I3) already exclude the range $m \leq 49$. Indeed, the function

$$U(m) = 1.4784 m \delta^m$$

is strictly increasing for positive integers m , since $\delta > 1$, and the direct numerical comparison

$$U(49) = 1.4784 \cdot 49 \cdot (\log_2 3)^{49} < 7 \cdot 10^{11}$$

shows that $K > 7 \cdot 10^{11}$ and $K < U(m)$ are incompatible for $m \leq 49$. Hence every remaining hypothetical non-trivial m -cycle considered below has $m \geq 50$.

All other estimates and reductions used below are proved in the paper or reduced to exact finite checks. In particular, we do not use Hercher's Theorem 21 [2]; the suffix-balanced argument below is used as a replacement for the consecutive-block contribution estimate.

3. Elementary Cycle Identities

The following elementary transition identity is standard in the study of Collatz m -cycles; see, for example, [6, 2].

Lemma 1 (Local-minimum transition). *Let n_i be a local minimum, and let k_i and ℓ_i be the following odd-run and even-run lengths. Then*

$$n_i = a_i 2^{k_i} - 1$$

for some odd integer a_i , and

$$n_{i+1} = \frac{a_i 3^{k_i} - 1}{2^{\ell_i}}.$$

Equivalently,

$$a_i 3^{k_i} - 1 \equiv 2^{\ell_i} \pmod{2^{\ell_i+1}}.$$

Proof. Since the odd run has exact length k_i , we have $n_i + 1 = a_i 2^{k_i}$ with a_i odd. Repeated use of the odd branch gives

$$C^t(n_i) = a_i 3^t 2^{k_i-t} - 1 \quad (0 \leq t \leq k_i).$$

In particular,

$$C^{k_i}(n_i) = a_i 3^{k_i} - 1.$$

The following ℓ_i steps are exactly even divisions and end at n_{i+1} . Hence

$$a_i 3^{k_i} - 1 = 2^{\ell_i} n_{i+1},$$

which gives the transition formula.

Finally, n_{i+1} is odd, say $n_{i+1} = 2r + 1$. Therefore

$$a_i 3^{k_i} - 1 = 2^{\ell_i} (2r + 1) \equiv 2^{\ell_i} \pmod{2^{\ell_i+1}}.$$

Conversely, this congruence says that division by 2^{ℓ_i} leaves an odd quotient, so the even run has exact length ℓ_i . $\square$

The next inequality is the standard product estimate for Collatz m -cycles, written in our notation; compare Hercher [2, Theorem 16].

Lemma 2 (Product inequality). *For every non-trivial cycle,*

$$\delta < \frac{K+L}{K} < \delta + \frac{1}{3K \log 2} \sum_{i=1}^m T(n_i).$$

Proof. Multiplying all step ratios around the cycle gives

$$1 = \frac{3^K}{2^{K+L}} \prod_{\text{odd steps } x} \left(1 + \frac{1}{3x}\right).$$

Taking logarithms,

$$(K + L) \log 2 - K \log 3 = \sum_{\text{odd steps } x} \log \left(1 + \frac{1}{3x}\right) > 0.$$

This gives the lower bound. Since $\log(1 + u) < u$ for $u > 0$,

$$(K + L) \log 2 - K \log 3 < \sum_{\text{odd steps } x} \frac{1}{3x}.$$

The odd steps are exactly $C^t(n_i)$, where $1 \leq i \leq m$ and $0 \leq t < k_i$. Hence

$$\sum_{\text{odd steps } x} \frac{1}{x} = \sum_{i=1}^m T(n_i).$$

Dividing by $K \log 2$ proves the upper bound. $\square$

The following elementary estimate is Hercher's standard bound for $T(n_i)$; see [2, Remark 7]. We recall the short proof for completeness.

Lemma 3 (Elementary bound for T). *For every local minimum n ,*

$$T(n) < \frac{3}{n}.$$

Proof. During the following odd run, each odd step satisfies

$$C(x) = \frac{3x + 1}{2} > \frac{3}{2}x.$$

Hence

$$C^t(n) \geq \left(\frac{3}{2}\right)^t n \quad (0 \leq t < k),$$

with strict inequality for $t \geq 1$. Therefore

$$T(n) = \sum_{t=0}^{k-1} \frac{1}{C^t(n)} \leq \frac{1}{n} \sum_{t=0}^{k-1} \left(\frac{2}{3}\right)^t < \frac{1}{n} \sum_{t=0}^{\infty} \left(\frac{2}{3}\right)^t = \frac{3}{n}.$$

$\square$

The following growth estimate is standard; compare Hercher [2, Lemma 20 and the proof of Theorem 21].

Lemma 4 (Growth of consecutive local minima). *For consecutive local minima in a non-trivial cycle,*

$$n_{i+1} + 1 < (n_i + 1)^\delta.$$

Moreover, for every integer $j \geq 0$,

$$k_{i+j} \leq \delta^j \log_2(n_i + 1),$$

and the inequality is strict when $j \geq 1$.

Proof. Write $n_i = a2^{k_i} - 1$. By Lemma 1,

$$n_{i+1} + 1 = \frac{a3^{k_i} - 1}{2^{\ell_i}} + 1.$$

Since $\ell_i \geq 1$ and $a3^{k_i} \geq 3$,

$$n_{i+1} + 1 < \frac{a3^{k_i}}{2} + 1 \leq a3^{k_i}.$$

As $a \geq 1$ and $\delta = \log_2 3 > 1$,

$$a3^{k_i} \leq a^\delta 3^{k_i} = (a2^{k_i})^\delta = (n_i + 1)^\delta.$$

This proves

$$n_{i+1} + 1 < (n_i + 1)^\delta.$$

Also $2^{k_i} \leq n_i + 1$, so

$$k_i \leq \log_2(n_i + 1).$$

Iterating the first inequality gives, for $j \geq 1$,

$$\log_2(n_{i+j} + 1) < \delta^j \log_2(n_i + 1).$$

Together with $k_{i+j} \leq \log_2(n_{i+j} + 1)$, this proves the stated bounds. $\square$

The following is only the summed form of the preceding growth estimate; it is the form needed later for suffix-balanced blocks.

Lemma 5 (Block form of the growth estimate). *Let $n_r, n_{r+1}, \dots, n_{r+s-1}$ be s consecutive local minima, with indices read cyclically, and put*

$$K_{r,s} = \sum_{j=0}^{s-1} k_{r+j}.$$

Then

$$K_{r,s} \leq \log_2(n_r + 1) \frac{\delta^s - 1}{\delta - 1}.$$

Consequently,

$$n_r + 1 \geq 2^{K_{r,s}(\delta-1)/(\delta^s-1)}.$$

Proof. By Lemma 4,

$$k_{r+j} \leq \delta^j \log_2(n_r + 1) \quad (0 \leq j \leq s-1).$$

Summing these s inequalities gives

$$\begin{aligned} K_{r,s} &\leq \log_2(n_r + 1) \sum_{j=0}^{s-1} \delta^j \\ &= \log_2(n_r + 1) \frac{\delta^s - 1}{\delta - 1}. \end{aligned}$$

Rearranging this inequality gives the stated lower bound for $n_r + 1$. $\square$

4. A Suffix-Balanced Block Selection

The next lemma is an elementary fact about cyclic sums. We first cut the cycle at an arbitrary point and then choose a highest partial-sum point as the end of the block. This guarantees that every suffix of the chosen block has at least its average share of the total odd-run length.

Lemma 6 (Suffix-balanced consecutive block). *Let $k_1, \dots, k_m$ be the odd-run lengths, read cyclically, and let*

$$K = \sum_{i=1}^m k_i.$$

For every s with $1 \leq s \leq m$, there is a consecutive block of s local minima which, after cyclic relabelling as

$$B = (n_1, n_2, \dots, n_s),$$

satisfies

$$\sum_{j=s-t+1}^s k_j \geq \frac{tK}{m} \quad (1 \leq t \leq s).$$

Here k_j denotes the odd-run length following the relabelled local minimum n_j .

Proof. Choose an arbitrary point of the cycle and write the cyclic sequence as $k_1, \dots, k_m$. Put

$$\mu = \frac{K}{m}, \quad b_i = k_i - \mu.$$

Then

$$\sum_{i=1}^m b_i = 0.$$

Define

$$A_0 = 0, \quad A_r = \sum_{i=1}^r b_i \quad (1 \leq r \leq m).$$

Since $A_m = 0$, choose an index $e \in \{1, \dots, m\}$ such that A_e is maximal among $A_0, A_1, \dots, A_m$. Take the consecutive block of length s ending at e , with indices read cyclically.

We claim that every suffix of this block has nonnegative adjusted sum. Let $1 \leq t \leq s$. If the suffix of length t does not cross the initial cut, that is, if $t \leq e$, then its adjusted sum is

$$b_{e-t+1} + \dots + b_e = A_e - A_{e-t} \geq 0,$$

by the maximality of A_e .

If $t > e$, the suffix wraps around the initial cut. Its adjusted sum is

$$(b_{m+e-t+1} + \dots + b_m) + (b_1 + \dots + b_e).$$

Using $A_m = 0$, this equals

$$(A_m - A_{m+e-t}) + A_e = A_e - A_{m+e-t} \geq 0,$$

again by the maximality of A_e .

Thus every suffix of the chosen block satisfies

$$\sum (k_i - \mu) \geq 0.$$

Equivalently, every suffix of length t satisfies

$$\sum k_i \geq t\mu = \frac{tK}{m}.$$

Relabelling the chosen block as $n_1, \dots, n_s$ gives

$$\sum_{j=s-t+1}^s k_j \geq \frac{tK}{m} \quad (1 \leq t \leq s),$$

as required. □

Lemma 7 (Residual estimate for a suffix-balanced block). *Let $1 \leq s \leq m$, and let*

$$B = (n_1, \dots, n_s)$$

be a suffix-balanced block as in Lemma 6. Let R be the set of the $m - s$ local minima not in B . Suppose that

$$\sum_{n_i \in R} T(n_i) < \rho.$$

For $1 \leq t \leq s$, define

$$V_t = \frac{tK}{m} \frac{\delta - 1}{\delta^t - 1}.$$

Then

$$\delta < \frac{K + L}{K} < \delta + \frac{1}{3K \log 2} \left(\rho + \sum_{t=1}^s \frac{3}{2^{V_t} - 1} \right).$$

Proof. By Lemma 2, it remains to give an upper bound for

$$\sum_{i=1}^m T(n_i).$$

The local minima outside B contribute less than ρ by assumption.

We now bound the contribution from B . Fix $j \in \{1, \dots, s\}$, and put

$$t = s - j + 1.$$

Then

$$n_j, n_{j+1}, \dots, n_s$$

is a suffix of B of length t . Since B is suffix-balanced,

$$\sum_{r=j}^s k_r \geq \frac{tK}{m}.$$

Applying Lemma 5 to this suffix gives

$$n_j + 1 \geq 2^{(tK/m)(\delta-1)/(\delta^t-1)} = 2^{V_t}.$$

Therefore

$$n_j \geq 2^{V_t} - 1.$$

By Lemma 3,

$$T(n_j) < \frac{3}{n_j} \leq \frac{3}{2^{V_t} - 1}.$$

Summing this estimate over $j = 1, \dots, s$, equivalently over $t = s, \dots, 1$, we obtain

$$\sum_{n_i \in B} T(n_i) < \sum_{t=1}^s \frac{3}{2^{V_t} - 1}.$$

Together with the omitted contribution, this gives

$$\sum_{i=1}^m T(n_i) < \rho + \sum_{t=1}^s \frac{3}{2^{V_t} - 1}.$$

Substituting this bound into Lemma 2 proves the claim. $\square$

Remark 8. The term $3/(2^{V_t} - 1)$ is a computable substitute for $3/n_j$. The actual value of n_j is unknown, but the suffix-balanced block and Lemma 5 give the certified lower bound

$$n_j \geq 2^{V_t} - 1.$$

If another lower bound $n_j > X_0$ is available, the corresponding contribution may be bounded by

$$\min \left\{ \frac{3}{X_0}, \frac{3}{2^{V_t} - 1} \right\}.$$

Lemma 9 (Monotonicity of suffix exponents). *Let $\delta = \log_2 3$. If $1 \leq t \leq s$, then*

$$\frac{t(\delta - 1)}{\delta^t - 1} \geq \frac{s(\delta - 1)}{\delta^s - 1}.$$

Consequently, for fixed K and m ,

$$V_t = \frac{tK}{m} \frac{\delta - 1}{\delta^t - 1}$$

satisfies $V_t \geq V_s$ for $1 \leq t \leq s$.

Proof. It suffices to show that $g(t) = t/(\delta^t - 1)$ is decreasing for integers $t \geq 1$. The inequality $g(t+1) < g(t)$ is equivalent to

$$\delta^t((t+1) - t\delta) < 1.$$

For $t = 1$, this is $\delta(2 - \delta) < 1$, since

$$\delta(2 - \delta) = 1 - (\delta - 1)^2 < 1.$$

For $t \geq 2$, using $\delta > 3/2$, we have

$$(t+1) - t\delta \leq 1 - \frac{t}{2} \leq 0,$$

so the same inequality is immediate. Hence g is decreasing, and the claims follow. $\square$

Lemma 10 (Continued-fraction denominator test). *Let $\epsilon > 0$. Suppose that the interval $(\delta, \delta + \epsilon)$ contains no rational number whose reduced denominator is smaller than Q . If*

$$\delta < \frac{K+L}{K} < \delta + \epsilon,$$

then $K \geq Q$.

Proof. This is the standard continued-fraction separation principle; compare, for example, [3, Ch. I]. The number $(K+L)/K$ is rational and, after reduction, has denominator dividing K . If $K < Q$, this rational number would have reduced denominator smaller than Q and would lie in $(\delta, \delta + \epsilon)$, contrary to the hypothesis. $\square$

5. A Uniform Lower-Bound Iteration

This section turns the global lower bound for the total number of odd steps into a sequence of stronger lower bounds. The argument is not a search for Collatz segments. Each step has the following form:

$$K \geq A \quad \implies \quad K \geq B.$$

The implication is obtained by bounding the product-error term in Lemma 2 and then applying the continued-fraction separation test of Lemma 10. The numerical comparisons used in this section are exact rational-interval comparisons.

Lemma 11 (One step of the lower-bound iteration). *Let $m_* \geq 1$, let $1 \leq s \leq m_*$, and let $A > 0$. For $1 \leq t \leq s$, put*

$$V_t(A, m_*) = \frac{tA}{m_*} \frac{\delta - 1}{\delta^t - 1}.$$

Define

$$E_s(A, m_*) = \frac{3(m_* - s)}{X} + \sum_{t=1}^s \frac{3}{2^{V_t(A, m_*)} - 1}$$

and

$$\epsilon_s(A, m_*) = \frac{E_s(A, m_*)}{3A \log 2}.$$

Suppose that the interval

$$(\delta, \delta + \epsilon_s(A, m_*))$$

contains no rational number whose reduced denominator is smaller than B . Then, for every m with $s \leq m \leq m_$, every non-trivial m -cycle with $K \geq A$ satisfies*

$$K \geq B.$$

Proof. Fix a hypothetical m -cycle with $s \leq m \leq m_*$ and $K \geq A$. By Lemma 6, choose a suffix-balanced block

$$B_s = (n_1, \dots, n_s).$$

Let R be the set of local minima outside this block. Since R has $m - s$ elements, Input (I1) and Lemma 3 give

$$\sum_{n_i \in R} T(n_i) < \frac{3(m-s)}{X} \leq \frac{3(m_*-s)}{X}.$$

It remains to bound the contribution from the selected block. Fix $j \in \{1, \dots, s\}$ and set $t = s - j + 1$. The suffix

$$n_j, n_{j+1}, \dots, n_s$$

has length t . By suffix-balance,

$$\sum_{r=j}^s k_r \geq \frac{tK}{m}.$$

Since $K \geq A$ and $m \leq m_*$, we have

$$\frac{tK}{m} \geq \frac{tA}{m_*}.$$

Lemma 5, applied to this suffix, therefore gives

$$n_j + 1 \geq 2^{(tK/m)(\delta-1)/(\delta^t-1)} \geq 2^{V_t(A, m_*)}.$$

Consequently,

$$T(n_j) < \frac{3}{n_j} \leq \frac{3}{2^{V_t(A, m_*)} - 1}.$$

Summing over $j = 1, \dots, s$, equivalently over $t = 1, \dots, s$, gives

$$\sum_{i=1}^m T(n_i) < E_s(A, m_*).$$

Substituting this estimate into Lemma 2, and using $K \geq A$, yields

$$\delta < \frac{K+L}{K} < \delta + \frac{E_s(A, m_*)}{3K \log 2} \leq \delta + \epsilon_s(A, m_*).$$

The rational number $(K+L)/K$, after reduction, has denominator dividing K . If $K < B$, this reduced denominator would be smaller than B , contradicting the assumed continued-fraction exclusion. Hence $K \geq B$. $\square$

Lemma 12 (Certified lower-bound iteration). *Put*

$$K^{(0)} = 700000000000.$$

The following rows are certified by exact rational-interval arithmetic. In row j , Lemma 11 is applied with

$$A = K^{(j-1)}, \quad s = s_j, \quad m_* = 95.$$

j	s_j	$K^{(j)}$
1	47	5750934602875680
2	67	397560349370386783
3	77	4640282259296926456
4	82	27444133206411171953
5	86	77692117359936589403
6	88	205632218873398596256

For each $j = 1, \dots, 6$, the certified implication is

$$K \geq K^{(j-1)} \quad \implies \quad K \geq K^{(j)}$$

for every actual m -cycle with

$$s_j \leq m \leq 95.$$

Thus the rows are not used all at once. In Lemma 13, they are invoked one after another, and before row j is used the previous rows have already reduced the current range of possible m -values enough to guarantee $s_j \leq m$ throughout that current range. We write

$$K_0 = K^{(6)} = 205632218873398596256.$$

In the special case $m = 91$, after the six rows have given $K \geq K_0$, a separate full-block continued-fraction calculation certifies

$$K \geq Q, \quad Q = 7941964418702608664581.$$

Proof. Fix $j \in \{1, \dots, 6\}$. The calculation computes

$$\epsilon_{s_j}(K^{(j-1)}, 95)$$

and verifies that

$$\left(\delta, \delta + \epsilon_{s_j}(K^{(j-1)}, 95) \right)$$

contains no rational number whose reduced denominator is smaller than $K^{(j)}$. Lemma 11 therefore gives the displayed implication for every m satisfying $s_j \leq m \leq 95$.

For $m = 91$, the final full-block calculation uses

$$s = m = m_* = 91, \quad A = K_0.$$

The calculation computes $\epsilon_{91}(K_0, 91)$ and verifies that

$$(\delta, \delta + \epsilon_{91}(K_0, 91))$$

contains no rational number whose reduced denominator is smaller than Q . Lemma 10 then gives $K \geq Q$. $\square$

Lemma 13 (Exclusion of $m \leq 91$). *Assuming Inputs (I2) and (I3), there is no non-trivial Collatz m -cycle with $m \leq 91$.*

Proof. Let

$$U(x) = 1.4784x\delta^x.$$

Since $\delta > 1$, the function U is strictly increasing for $x > 0$. For a positive number B , let $m^*(B)$ denote the unique real solution of

$$U(x) = B.$$

If a hypothetical m -cycle satisfies $K \geq B$, then the Simons–de Weger upper bound $K < U(m)$ implies

$$U(m) > B.$$

By monotonicity, this gives the necessary condition

$$m > m^*(B).$$

Consequently, once we know $a < m^*(B) < a + 1$, every surviving integer m must satisfy $m \geq a + 1$.

The certified root intervals used below are

lower bound B	certified interval for $m^*(B)$	integer consequence
$K^{(0)}$	$49 < m^*(K^{(0)}) < 50$	$m \geq 50$
$K^{(1)}$	$68 < m^*(K^{(1)}) < 69$	$m \geq 69$
$K^{(2)}$	$77 < m^*(K^{(2)}) < 78$	$m \geq 78$
$K^{(3)}$	$82 < m^*(K^{(3)}) < 83$	$m \geq 83$
$K^{(4)}$	$86 < m^*(K^{(4)}) < 87$	$m \geq 87$
$K^{(5)}$	$88 < m^*(K^{(5)}) < 89$	$m \geq 89$
$K^{(6)}$	$90 < m^*(K^{(6)}) < 91$	$m \geq 91$
Q	$98 < m^*(Q) < 99$	$m \geq 99$.

The initial lower bound $K > K^{(0)}$ gives $m \geq 50$, so $m \leq 49$ is excluded. We now use the six rows of Lemma 12 as a single ordered chain. The point is that the next row is invoked only after the preceding rows have shortened the current range of possible m -values. Hence the recorded block length s_j covers every m still under consideration when row j is used.

The induction is:

step	current range	s_j	new condition	excluded values
1	$50 \leq m \leq 91$	$47 \leq 50$	$m \geq 69$	$50 \leq m \leq 68$
2	$69 \leq m \leq 91$	$67 \leq 69$	$m \geq 78$	$69 \leq m \leq 77$
3	$78 \leq m \leq 91$	$77 \leq 78$	$m \geq 83$	$78 \leq m \leq 82$
4	$83 \leq m \leq 91$	$82 \leq 83$	$m \geq 87$	$83 \leq m \leq 86$
5	$87 \leq m \leq 91$	$86 \leq 87$	$m \geq 89$	$87 \leq m \leq 88$
6	$89 \leq m \leq 91$	$88 \leq 89$	$m \geq 91$	$89 \leq m \leq 90$

Here is the first step in detail. The current range is $50 \leq m \leq 91$, and $s_1 = 47 \leq 50$, so row 1 is valid for every m in that range. It gives $K \geq K^{(1)}$. The root interval for $K^{(1)}$ then forces $m \geq 69$. Intersecting this necessary condition with the current range removes exactly $50 \leq m \leq 68$. The next current range is therefore $69 \leq m \leq 91$, and only then is row 2 invoked. The remaining rows follow in the same way, as displayed in the table. Thus all cases $m \leq 90$ are excluded.

It remains only to consider $m = 91$. For this value, Lemma 12 gives

$$K \geq Q = 7941964418702608664581.$$

The root interval for Q forces $m \geq 99$, which is incompatible with $m = 91$. This excludes $m = 91$, and the proof is complete. $\square$

6. Finite Exclusions by the Affine a -Ladder

This section gives the finite-search mechanism used for the cases $m = 92, 93, 94, 95$. The mechanism is as follows. We start from

$$n_1 = a_1 2^{k_1} - 1, \quad a = a_1,$$

and then use the recurrence for the quantities a_i . After the exponent prefix has been fixed, each later a_i , and therefore each later local minimum n_i , is an explicit expression in the same initial integer $a = a_1$. Conversely, each admissible value of a later local minimum gives an interval condition for this same a_1 , and each exact power-of-two condition gives a residue class for this same a_1 .

For a fixed prefix of length i , the surviving initial parameters form a set

$$\mathcal{A}_i = [L_i, U_i] \cap \{a_1 : a_1 \equiv r_i \pmod{M_i}\}.$$

The full level i set $\mathcal{S}_i$ is the union of these sets over all allowed exponent prefixes of length i . If a genuine segment existed, its true value of a_1 would survive at every checked level. Hence an empty level,

$$\mathcal{S}_i = \emptyset,$$

is already a contradiction. The verifier may therefore choose a convenient depth d and check directly that $\mathcal{S}_d$ is empty. This is what is done in the four finite cases below.

6.1. The Recurrence from a_i to a_{i+1}

For each local minimum we have

$$n_i = a_i 2^{k_i} - 1.$$

The following local-minimum transition is

$$n_{i+1} = \frac{a_i 3^{k_i} - 1}{2^{\ell_i}}.$$

The next local minimum also has the form

$$n_{i+1} = a_{i+1} 2^{k_{i+1}} - 1,$$

so

$$n_{i+1} + 1 = a_{i+1} 2^{k_{i+1}}.$$

Hence

$$a_{i+1} = \frac{n_{i+1} + 1}{2^{k_{i+1}}}.$$

Substituting the transition formula gives

$$\begin{aligned} a_{i+1} &= \frac{1}{2^{k_{i+1}}} \left(\frac{a_i 3^{k_i} - 1}{2^{\ell_i}} + 1 \right) \\ &= \frac{a_i 3^{k_i} + 2^{\ell_i} - 1}{2^{\ell_i + k_{i+1}}}. \end{aligned}$$

Thus the basic recurrence is

$$\boxed{a_{i+1} = \frac{a_i 3^{k_i} + 2^{\ell_i} - 1}{2^{\ell_i + k_{i+1}}}.} \tag{1}$$

This is the recurrence used in the finite checks. Once the exponents are fixed, the next a -value is a linear expression in the previous a -value, divided by a power of two.

6.2. Closed Form for a_i in Terms of a_1

Put

$$a = a_1.$$

For fixed exponents

$$k_1, \ell_1, k_2, \ell_2, \dots, k_{i-1}, \ell_{i-1}, k_i,$$

write

$$a_i = \frac{p_i a + q_i}{2^{s_i}}. \quad (2)$$

The initial values are

$$p_1 = 1, \quad q_1 = 0, \quad s_1 = 0.$$

Using (1), if

$$a_i = \frac{p_i a + q_i}{2^{s_i}},$$

then

$$a_{i+1} = \frac{3^{k_i}(p_i a + q_i) + (2^{\ell_i} - 1)2^{s_i}}{2^{s_i + \ell_i + k_{i+1}}}.$$

Therefore

$$\boxed{p_{i+1} = 3^{k_i} p_i}, \quad \boxed{q_{i+1} = 3^{k_i} q_i + (2^{\ell_i} - 1)2^{s_i}}, \quad \boxed{s_{i+1} = s_i + \ell_i + k_{i+1}}. \quad (3)$$

By induction,

$$p_i = 3^{k_1 + \dots + k_{i-1}},$$

$$s_i = \ell_1 + \dots + \ell_{i-1} + k_2 + \dots + k_i,$$

and

$$q_i = \sum_{h=1}^{i-1} (2^{\ell_h} - 1)2^{s_h} 3^{k_{h+1} + \dots + k_{i-1}}.$$

Here an empty sum is 0, and an empty product is 1. Hence

$$\boxed{a_i = \frac{3^{k_1 + \dots + k_{i-1}} a + \sum_{h=1}^{i-1} (2^{\ell_h} - 1)2^{s_h} 3^{k_{h+1} + \dots + k_{i-1}}}{2^{s_i}}}. \quad (4)$$

Thus every a_i is determined by the same initial integer $a = a_1$ and by the fixed exponent prefix.

6.3. Writing n_i in Terms of a_1

Since

$$n_i = a_i 2^{k_i} - 1,$$

formula (2) gives

$$n_i = \frac{2^{k_i}(p_i a + q_i) - 2^{s_i}}{2^{s_i}}. \quad (5)$$

Equivalently, applying the next transition directly gives

$$n_{i+1} = \frac{a_i 3^{k_i} - 1}{2^{\ell_i}} = \frac{3^{k_i}(p_i a + q_i) - 2^{s_i}}{2^{s_i + \ell_i}}. \quad (6)$$

Therefore, after the exponent prefix has been fixed, every n_i is an affine expression in the single variable $a = a_1$, divided by a power of two.

For instance,

$$n_2 = \frac{a 3^{k_1} - 1}{2^{\ell_1}}.$$

Since

$$n_2 + 1 = \frac{a 3^{k_1} + 2^{\ell_1} - 1}{2^{\ell_1}} = a_2 2^{k_2},$$

we have

$$a_2 = \frac{a 3^{k_1} + 2^{\ell_1} - 1}{2^{\ell_1 + k_2}}.$$

Substitution into

$$n_3 = \frac{a_2 3^{k_2} - 1}{2^{\ell_2}}$$

gives the two-step formula

$$n_3 = \frac{a 3^{k_1 + k_2} + (2^{\ell_1} - 1) 3^{k_2} - 2^{\ell_1 + k_2}}{2^{\ell_1 + k_2 + \ell_2}}. \quad (7)$$

This is the numerator used in the two-step case.

6.4. Solving Back for a_1 from a Layer

The same formulas may be read backwards. From

$$n_i + 1 = a_i 2^{k_i} = \frac{(p_i a + q_i) 2^{k_i}}{2^{s_i}},$$

we get

$$2^{s_i}(n_i + 1) = 2^{k_i}(p_i a + q_i).$$

Therefore

$$a = \frac{2^{s_i}(n_i + 1) - 2^{k_i}q_i}{2^{k_i}p_i}. \quad (8)$$

This is the first inverse form: a fixed layer value n_i , together with the exponent prefix, determines the same initial parameter $a = a_1$.

There is also a useful inverse form from the transition to n_{i+1} . From (6),

$$n_{i+1} = \frac{3^{k_i}(p_i a + q_i) - 2^{s_i}}{2^{s_i+\ell_i}},$$

so

$$2^{s_i+\ell_i}n_{i+1} = 3^{k_i}p_i a + 3^{k_i}q_i - 2^{s_i}.$$

Hence

$$a = \frac{2^{s_i+\ell_i}n_{i+1} + 2^{s_i} - 3^{k_i}q_i}{3^{k_i}p_i}. \quad (9)$$

This is the exact expression behind the interval cut. If

$$N_{i+1}^- \leq n_{i+1} \leq N_{i+1}^+,$$

then (9) gives immediately

$$a \geq \left\lceil \frac{N_{i+1}^- 2^{s_i+\ell_i} + 2^{s_i} - 3^{k_i}q_i}{3^{k_i}p_i} \right\rceil \quad (10)$$

and

$$a \leq \left\lfloor \frac{N_{i+1}^+ 2^{s_i+\ell_i} + 2^{s_i} - 3^{k_i}q_i}{3^{k_i}p_i} \right\rfloor. \quad (11)$$

Thus each layer gives another interval condition on the same initial parameter $a = a_1$.

6.5. The Range of k_i and ℓ_i

The range of k_i comes directly from the range of n_i . If

$$N_i^- \leq n_i \leq N_i^+,$$

then

$$2^{k_i} \leq n_i + 1 \leq N_i^+ + 1.$$

Hence

$$1 \leq k_i \leq \lfloor \log_2(N_i^+ + 1) \rfloor. \quad (12)$$

We next derive the range of ℓ_i without first forming a separate range for a_i . The two transition identities are

$$n_i = a_i 2^{k_i} - 1, \quad n_{i+1} = \frac{a_i 3^{k_i} - 1}{2^{\ell_i}}.$$

Eliminating a_i gives

$$a_i = \frac{n_i + 1}{2^{k_i}}$$

and hence

$$\boxed{3^{k_i}(n_i + 1) - 2^{k_i} = 2^{k_i + \ell_i} n_{i+1}.} \quad (13)$$

Equivalently,

$$3^{k_i}(n_i + 1) = 2^{k_i}(2^{\ell_i} n_{i+1} + 1).$$

Thus after k_i is fixed, the possible values of ℓ_i are controlled only by the ranges of n_i and n_{i+1} .

Suppose that

$$N_i^- \leq n_i \leq N_i^+, \quad n_{i+1} \geq N_{i+1}^-.$$

From (13) we get

$$2^{k_i + \ell_i} N_{i+1}^- \leq 3^{k_i}(N_i^+ + 1) - 2^{k_i}.$$

Therefore

$$2^{\ell_i} \leq \frac{3^{k_i}(N_i^+ + 1) - 2^{k_i}}{2^{k_i} N_{i+1}^-}.$$

Put

$$T_i = \left\lfloor \frac{3^{k_i}(N_i^+ + 1) - 2^{k_i}}{2^{k_i} N_{i+1}^-} \right\rfloor. \quad (14)$$

If $T_i < 2$, then no $\ell_i \geq 1$ is possible. Otherwise every genuine branch satisfies

$$\boxed{1 \leq \ell_i \leq \lfloor \log_2 T_i \rfloor.} \quad (15)$$

This is the finite range of ℓ_i used in the computation. The point is that ℓ_i is bounded directly from k_i and the two local-minimum ranges. No intermediate range for a_i is needed in this step.

If an upper bound $n_{i+1} \leq N_{i+1}^+$ is also available, the same identity can also give a lower bound for ℓ_i . Namely,

$$2^{k_i + \ell_i} N_{i+1}^+ \geq 3^{k_i}(N_i^- + 1) - 2^{k_i},$$

so one may impose

$$2^{\ell_i} \geq \frac{3^{k_i}(N_i^- + 1) - 2^{k_i}}{2^{k_i} N_{i+1}^+}$$

whenever this improves the trivial lower bound $\ell_i \geq 1$. In the finite checks below, the upper bound (15) is the essential one.

6.6. The Congruence Filter for a_1

The interval conditions above only restrict the size of a . The exact powers of two give residue classes for the same a . Define

$$B_i(a) = 3^{k_i}(p_i a + q_i) - 2^{s_i}.$$

By (6),

$$n_{i+1} = \frac{B_i(a)}{2^{s_i+\ell_i}}.$$

The exact even-run condition is

$$v_2(B_i(a)) = s_i + \ell_i.$$

Equivalently,

$$B_i(a) \equiv 2^{s_i+\ell_i} \pmod{2^{s_i+\ell_i+1}}.$$

Since $3^{k_i}p_i$ is odd, it is invertible modulo this power of two. Hence

$$\boxed{a \equiv \left(2^{s_i+\ell_i} + 2^{s_i} - 3^{k_i}q_i\right) (3^{k_i}p_i)^{-1} \pmod{2^{s_i+\ell_i+1}}.} \quad (16)$$

If k_{i+1} has also been fixed, we use the stronger condition that the next odd-run length is exact. From

$$a_{i+1} = \frac{p_{i+1}a + q_{i+1}}{2^{s_{i+1}}}, \quad s_{i+1} = s_i + \ell_i + k_{i+1},$$

the condition $v_2(n_{i+1} + 1) = k_{i+1}$ is

$$v_2(p_{i+1}a + q_{i+1}) = s_{i+1}.$$

Therefore

$$p_{i+1}a + q_{i+1} \equiv 2^{s_{i+1}} \pmod{2^{s_{i+1}+1}},$$

and hence

$$\boxed{a \equiv (2^{s_{i+1}} - q_{i+1})p_{i+1}^{-1} \pmod{2^{s_{i+1}+1}}.} \quad (17)$$

This residue class is precisely the way in which the ℓ_i -condition is used. We do not keep a separate unrelated value of a_i . The ℓ_i -choice is converted into a congruence for the same initial parameter $a = a_1$.

6.7. The Direct Depth- d Test

The verifier is a direct test at a chosen depth. It is not a collection of independent one-layer tests for unrelated values $a_1, a_2, a_3, \dots$. All conditions are pulled back to the same initial integer

$$a = a_1.$$

For a fixed exponent prefix of length i , the live set has the form

$$\mathcal{A}_i(k_1, \ell_1, \dots, k_i, \ell_i) = [L_i, U_i] \cap \{a : a \equiv r_i \pmod{M_i}\},$$

where M_i is a power of two. This set contains exactly the initial parameters $a = a_1$ which survive all size bounds and exact divisibility conditions imposed along that fixed prefix.

The full level- i candidate set is

$$\mathcal{S}_i = \bigcup_{(k_1, \ell_1, \dots, k_i, \ell_i)} \mathcal{A}_i(k_1, \ell_1, \dots, k_i, \ell_i),$$

where the union is over the finite exponent prefixes allowed by the ranges above. If a genuine Collatz segment satisfying the assumed bounds existed, then its true initial parameter a_1 and its true exponent prefix would give

$$a_1 \in \mathcal{S}_i$$

for every checked level i . Consequently,

$$\boxed{\mathcal{S}_i = \emptyset \text{ for some } i \implies \text{the assumed segment does not exist.}}$$

In practice one need not materialize $\mathcal{S}_1, \mathcal{S}_2, \dots$ as separate large lists. One may choose a depth d , enumerate all exponent prefixes of length d , and check directly whether

$$\mathcal{S}_d = \emptyset.$$

This is a single depth- d certificate. If an earlier level is already empty, the computation may stop earlier, but that is only an optimization. The proof only needs one certified empty level.

For each fixed prefix the update is as follows. The search maintains

$$\boxed{a \in [L, U] \cap \{a : a \equiv r \pmod{M}\}}.$$

Initially, after k_1 has been fixed and $N_1^- \leq n_1 \leq N_1^+$ is assumed,

$$L = \left\lceil \frac{N_1^- + 1}{2^{k_1}} \right\rceil, \quad U = \left\lfloor \frac{N_1^+ + 1}{2^{k_1}} \right\rfloor,$$

and

$$a \equiv 1 \pmod{2}.$$

The following operations are then applied along the chosen prefix.

- (1) Use the range of n_i to obtain the finite range of k_i from (12).
- (2) For a fixed k_i , combine the two transition identities and eliminate a_i . This gives (13), an equation involving only $k_i, \ell_i, n_i, n_{i+1}$. Use the ranges of n_i and n_{i+1} to obtain the finite range of ℓ_i from (14)–(15).
- (3) For a fixed ℓ_i , use the inverse formula (9) to convert

$$N_{i+1}^- \leq n_{i+1} \leq N_{i+1}^+$$

into a new interval for the same initial parameter $a = a_1$.

- (4) Use the exact divisibility congruence (16), or the stronger next-odd-run congruence (17) when k_{i+1} is fixed, to obtain a residue class for the same $a = a_1$.
- (5) Intersect the interval and residue class. If the current class is

$$a \equiv r \pmod{M},$$

then the least element in this class not smaller than L is

$$a_0 = r + M \left\lceil \frac{L - r}{M} \right\rceil.$$

The branch is alive exactly when

$$a_0 \leq U.$$

Thus the computation is a direct check of a single initial integer a_1 through all layers of the chosen prefix. Different layers do not supply independent unrelated values of a_i . Each layer supplies another interval condition or residue condition for the same $a = a_1$. The finite exclusion succeeds when the selected level $\mathcal{S}_d$ is empty.

The four finite exclusions use the following depths and initial windows:

case	d	assumed range for $n_{\min}$	k_1 range
$m = 92$	2	$X \leq n_{\min} \leq (73/10)X$	$1 \leq k_1 \leq 73$
$m = 93$	3	$X \leq n_{\min} \leq 15X$	$1 \leq k_1 \leq 74$
$m = 94$	5	$X \leq n_{\min} \leq 24X$	$1 \leq k_1 \leq 75$
$m = 95$	6	$X \leq n_{\min} \leq 24X$	$1 \leq k_1 \leq 75$

For each case, the finite certificate begins with an exact list of integer intervals

$$N_i^- \leq n_i \leq N_i^+$$

and the corresponding finite ranges for the odd-run lengths k_i . The case sections below record these exact range certificates. Because some upper endpoints are very large, the values are displayed vertically with line-breakable digits to avoid over-wide three-column tables. Endpoints that are naturally expressed in terms of X are displayed in that form, for example X , $15X$, $24X$, or $\lfloor (73/10)X \rfloor$. Once these windows are fixed, the dependent ℓ_i -ranges are computed, all size constraints are pulled back to the same initial parameter a_1 , and the exact residue classes are intersected. The finite exclusion is the certified statement

$$\mathcal{S}_d = \emptyset$$

for the listed depth d .

7. The Case $m = 92$

We now exclude the case $m = 92$. By Lemma 12, every hypothetical 92-cycle satisfies

$$K \geq K_0 = 205632218873398596256.$$

The proof is arranged in two steps. In Subsection 7.1, we derive a finite system of necessary conditions for the first two local-minimum transitions and show that this system has no integer solution. This rules out

$$X \leq n_{\min} \leq \frac{73}{10}X.$$

In Subsection 7.2, the new information $n_{\min} > (73/10)X$ is inserted back into the product inequality. The resulting rational interval around $\delta = \log_2 3$ is small enough that the continued-fraction denominator test contradicts the Simons–de Weger upper bound for K .

7.1. Exclusion of the Range $X \leq n_{\min} \leq (73/10)X$

Assume, for contradiction inside this subsection, that

$$X \leq n_{\min} \leq \frac{73}{10}X.$$

Rotate the cycle so that $n_1 = n_{\min}$, and write

$$n_1 = a_1 2^{k_1} - 1.$$

We check the two-step prefix

$$n_1 \longrightarrow n_2 \longrightarrow n_3,$$

so the depth of the finite exclusion is $d = 2$. The check uses the affine a_1 -ladder of Section 6, as follows.

certified ranges for n_i and k_i . Write the certified size windows as

$$N_i^- \leq n_i \leq N_i^+.$$

For the initial window put

$$N_1^- = X, \quad N_1^+ = \left\lfloor \frac{73}{10} X \right\rfloor.$$

Since

$$2^{k_1} \leq n_1 + 1 \leq N_1^+ + 1 < 2^{74},$$

the first odd-run range is

$$1 \leq k_1 \leq k_1^+ = 73.$$

The upper endpoint for the next local minimum is obtained from the endpoint value k_1^+ . The local transition gives

$$n_2 + 1 < (n_1 + 1) \left(\frac{3}{2} \right)^{k_1}.$$

Hence every admissible branch satisfies

$$n_2 + 1 < (N_1^+ + 1) \left(\frac{3}{2} \right)^{k_1^+},$$

and the certificate may take

$$N_2^+ = \left\lceil \frac{(N_1^+ + 1)3^{k_1^+}}{2^{k_1^+}} \right\rceil - 1 = 123342987508541979904829103131129490.$$

The lower endpoint is supplied by the certified suffix lower target for this layer. The endpoint used in the certificate is

$$N_2^- = 57049502556188976010416.$$

Applying (12) to the window $[N_2^-, N_2^+]$ gives

$$1 \leq k_2 \leq k_2^+ = 116.$$

Repeating the same endpoint construction with N_2^+ and $k_2^+ = 116$ gives

$$N_3^+ = \left\lceil \frac{(N_2^+ + 1)3^{k_2^+}}{2^{k_2^+}} \right\rceil - 1$$

Thus,

$$N_3^+ = 32938249414827033236487808628302479917874121989731037035.$$

The depth-two check imposes the certified terminal lower target

$$N_3^- = 1168976708683560900305353752529965856.$$

Thus the Stage-1 data for $m = 92$ are:

$$i = 1. \ N_1^- = X$$

$$N_1^+ = \left\lfloor \frac{73}{10} X \right\rfloor$$

$$1 \leq k_1 \leq 73.$$

$$i = 2. \ N_2^- = 57049502556188976010416$$

$$N_2^+ = 123342987508541979904829103131129490$$

$$1 \leq k_2 \leq 116.$$

$$i = 3. \ N_3^- = 1168976708683560900305353752529965856$$

$$N_3^+ = 32938249414827033236487808628302479917874121989731037035$$

terminal layer.

Dependent ranges for ℓ_i . After a value of k_i has been fixed, the corresponding even-run range is computed from the eliminated transition identity (13), together with (14)–(15). Thus ℓ_i is not a case-independent global variable. For each admitted k_i , this gives a dependent range

$$\ell_i^-(k_i) \leq \ell_i \leq \ell_i^+(k_i)$$

from the certified ranges of n_i and n_{i+1} .

Pulling size constraints back to a_1 . For each fixed exponent prefix

$$(k_1, \ell_1, k_2, \ell_2),$$

all size constraints are converted into intervals for the same initial integer a_1 . Both inverse forms (8) and (9) are used. Each formula gives an interval for a_1 , and the working interval is their intersection. This is the point at which the bounds for n_1, n_2, n_3 are all imposed on one and the same variable a_1 .

Exact 2-adic residue filtering. The exact powers of two are then imposed as congruence conditions on the same a_1 . The even-run condition is (16); when the next odd-run length is fixed, the next odd-run condition is (17). The two congruences are intersected as residue classes modulo powers of two. Hence each complete branch is reduced to one condition of the form

$$[L, U] \cap \{a_1 : a_1 \equiv r \pmod{2^b}\}.$$

The branch is alive exactly when this interval contains an integer in the residue class.

The enumeration covers all

$$1 \leq k_1 \leq 73, \quad 1 \leq k_2 \leq 116,$$

and all dependent ℓ_1, ℓ_2 -ranges. After the interval and residue conditions are intersected, the depth-two candidate set is empty:

$$\mathcal{S}_2 = \emptyset.$$

This contradicts the assumed range $X \leq n_{\min} \leq (73/10)X$. Consequently,

$$n_{\min} > \frac{73}{10}X.$$

7.2. Final Contradiction from $n_{\min} > (73/10)X$

We now use the new lower bound. Since $n_{\min}$ is the least local minimum, every local minimum satisfies

$$n_i > \frac{73}{10}X.$$

By Lemma 3,

$$T(n_i) < \frac{3}{n_i} < \frac{3}{(73/10)X} = \frac{30}{73X}.$$

Choose a suffix-balanced block

$$B = (n_1, \dots, n_{91})$$

of length 91. There are $92 - 91 = 1$ local minima outside this block. Let R be this one-element set. Its contribution is bounded by

$$\sum_{n_i \in R} T(n_i) < \frac{30}{73X}.$$

We now bound the contribution from the block. The first element of the block is treated separately because its suffix has length 91, whereas the other

block elements have suffix length at most 90. For the first term, Lemma 5 and suffix-balance give

$$n_1 + 1 \geq 2^{(91K/92)(\delta-1)/(\delta^{91}-1)} \geq 2^{(91K_0/92)(\delta-1)/(\delta^{91}-1)}.$$

An exact rational-arithmetic check gives

$$\frac{91K_0}{92} \frac{\delta-1}{\delta^{91}-1} > 74.7729, \quad 2^{74.7729} > 1.7087 \cdot 2^{74}.$$

Thus

$$n_1 > 1.7087 \cdot 2^{74} - 1.$$

Since

$$1.7087 \cdot 2^{74} = \frac{17086}{10000} 2^{74} + \frac{1}{10000} 2^{74} > \frac{17086}{10000} 2^{74} + 1,$$

we get

$$n_1 > \frac{17086}{10000} 2^{74}.$$

Therefore

$$T(n_1) < \frac{3}{n_1} < \frac{3 \cdot 10000}{17086 \cdot 2^{74}}.$$

For each of the remaining 90 block terms, let t be the suffix length starting at that term. Then $1 \leq t \leq 90$. The exponent obtained from the suffix-balanced block is

$$V_t = \frac{tK}{92} \frac{\delta-1}{\delta^t-1}.$$

Using $K \geq K_0$ and the monotonicity in Lemma 9,

$$V_t \geq \frac{90K_0}{92} \frac{\delta-1}{\delta^{90}-1}.$$

An exact rational-arithmetic check shows that the right side is larger than 117. Hence for each of these 90 terms,

$$n_j + 1 > 2^{117}, \quad n_j > 2^{117} - 1,$$

and so

$$T(n_j) < \frac{3}{2^{117}-1}.$$

The total contribution of these 90 terms is therefore less than

$$\frac{270}{2^{117}-1}.$$

Combining the omitted term, the first block term, and the remaining 90 block terms gives

$$\sum_{i=1}^{92} T(n_i) < E_{92},$$

where

$$E_{92} = \frac{30}{73X} + \frac{3 \cdot 10000}{17086 \cdot 2^{74}} + \frac{270}{2^{117} - 1}.$$

Substitute this estimate into Lemma 2. Since $K \geq K_0$,

$$\delta < \frac{K+L}{K} < \delta + \frac{E_{92}}{3K \log 2} \leq \delta + \frac{E_{92}}{3K_0 \log 2}.$$

An exact rational-arithmetic check gives

$$\frac{E_{92}}{3K_0 \log 2} < 6.2442 \cdot 10^{-43}.$$

Thus

$$\delta < \frac{K+L}{K} < \delta + 6.2442 \cdot 10^{-43}.$$

The number $(K+L)/K$ is rational. After reduction to lowest terms, its denominator divides K . The continued-fraction denominator calculation shows that the interval

$$(\delta, \delta + 6.2442 \cdot 10^{-43})$$

contains no rational number whose positive reduced denominator is smaller than

$$Q = 7941964418702608664581.$$

By Lemma 10, this forces

$$K \geq Q.$$

On the other hand, the Simons–de Weger upper bound gives

$$K < 1.4784 \cdot 92 \cdot \delta^{92},$$

and an exact rational-arithmetic check gives

$$1.4784 \cdot 92 \cdot \delta^{92} < Q.$$

Thus $K < Q$, contradicting $K \geq Q$. Hence no non-trivial 92-cycle exists.

8. The Case $m = 93$

We now exclude $m = 93$. By Lemma 12, every hypothetical 93-cycle satisfies

$$K \geq K_0 = 205632218873398596256.$$

Again the proof has two parts. Subsection 8.1 derives a finite three-step exclusion and proves

$$n_{\min} > 15X.$$

Subsection 8.2 uses this improved lower bound to obtain the final continued-fraction contradiction.

8.1. Exclusion of the Range $X \leq n_{\min} \leq 15X$

Assume, for contradiction inside this subsection, that

$$X \leq n_{\min} \leq 15X.$$

Rotate the cycle so that $n_1 = n_{\min}$, and write

$$n_1 = a_1 2^{k_1} - 1.$$

We use the same affine a_1 -ladder as in Subsection 7.1, now for the three-step prefix

$$n_1 \longrightarrow n_2 \longrightarrow n_3 \longrightarrow n_4, \quad d = 3.$$

Certified ranges for n_i and k_i . Write the certified size windows as

$$N_i^- \leq n_i \leq N_i^+.$$

For $m = 93$, the exact windows used by the certificate are:

$$\begin{aligned} i = 1. \quad N_1^- &= X \\ N_1^+ &= 15X \\ 1 &\leq k_1 \leq 74. \end{aligned}$$

$$\begin{aligned} i = 2. \quad N_2^- &= X \\ N_2^+ &= 380166742320848568199813229178661520 \\ 1 &\leq k_2 \leq 118. \end{aligned}$$

$$\begin{aligned} i = 3. \quad N_3^- &= 57049502556188974280503 \\ N_3^+ &= 228424503647345007890025253118709850350967115525589024296 \\ 1 &\leq k_3 \leq 187. \end{aligned}$$

$$\begin{aligned} i = 4. \quad N_4^- &= 1168976708683560810603026763323076753 \\ N_4^+ &= 1940028604167084613521955890519462882673683406023357803087399683917236474147 \\ &\quad 96574772980570 \\ &\quad \text{terminal layer.} \end{aligned}$$

Dependent ℓ_i -ranges, a_1 -intervals, and residue classes. For each fixed exponent prefix

$$(k_1, \ell_1, k_2, \ell_2, k_3, \ell_3),$$

the dependent ℓ_i -ranges are computed from (13). The size bounds for n_1, n_2, n_3, n_4 are converted into intersected intervals for the same a_1 , using (8) and (9). The two exact residue conditions (16) and (17) are then intersected. Thus each complete branch again ends as one interval and one residue class for a_1 .

The enumeration covers all $k_1 = 1, \dots, 74$, together with all dependent k_2, k_3 and ℓ_i -ranges. After the interval and residue conditions are intersected, the depth-three candidate set is empty:

$$\mathcal{S}_3 = \emptyset.$$

This contradicts the assumed range $X \leq n_{\min} \leq 15X$, and therefore

$$n_{\min} > 15X.$$

8.2. Final Contradiction from $n_{\min} > 15X$

Since $n_{\min} > 15X$, every local minimum satisfies $n_i > 15X$. Therefore Lemma 3 gives the uniform estimate

$$T(n_i) < \frac{3}{n_i} < \frac{3}{15X} = \frac{1}{5X}.$$

Choose a suffix-balanced block of length 90. A 93-cycle has 93 local minima, so there are $93 - 90 = 3$ local minima outside the block. Each of these contributes less than $1/(5X)$, and hence the omitted contribution is less than

$$\frac{3}{5X}.$$

Now consider a term inside the chosen block. If the suffix beginning at this term has length t , then $1 \leq t \leq 90$. Lemma 5 gives

$$n_j + 1 \geq 2^{V_t}, \quad V_t = \frac{tK}{93} \frac{\delta - 1}{\delta^t - 1}.$$

Using $K \geq K_0$ and Lemma 9,

$$V_t \geq \frac{90K_0}{93} \frac{\delta - 1}{\delta^{90} - 1}.$$

An exact rational-arithmetic check shows that this last quantity is larger than 115. Thus each of the 90 block terms satisfies

$$n_j > 2^{115} - 1, \quad T(n_j) < \frac{3}{2^{115} - 1}.$$

The whole block therefore contributes less than

$$\frac{270}{2^{115} - 1}.$$

Combining the omitted part and the block part gives

$$\sum_{i=1}^{93} T(n_i) < E_{93}, \quad E_{93} = \frac{3}{5X} + \frac{270}{2^{115} - 1}.$$

Substitution into Lemma 2 gives

$$\delta < \frac{K+L}{K} < \delta + \frac{E_{93}}{3K \log 2} \leq \delta + \frac{E_{93}}{3K_0 \log 2}.$$

An exact rational-arithmetic check gives

$$\frac{E_{93}}{3K_0 \log 2} < 5.943 \cdot 10^{-43}.$$

Hence

$$\delta < \frac{K+L}{K} < \delta + 5.943 \cdot 10^{-43}.$$

The continued-fraction denominator calculation shows that this interval contains no rational number whose positive reduced denominator is smaller than

$$Q = 7941964418702608664581.$$

The reduced denominator of $(K+L)/K$ divides K . Hence Lemma 10 forces $K \geq Q$. But the Simons–de Weger upper bound and an exact rational-arithmetic check give

$$K < 1.4784 \cdot 93 \cdot \delta^{93} < Q.$$

This contradiction excludes $m = 93$.

9. The Case $m = 94$

By Lemma 12, every hypothetical 94-cycle satisfies

$$K \geq K_0 = 205632218873398596256.$$

The proof again has two parts. Subsection 9.1 derives a five-step finite exclusion with ladder bounds and proves

$$n_{\min} > 24X.$$

Subsection 9.2 uses this bound to derive the final contradiction.

9.1. Exclusion of the Range $X \leq n_{\min} \leq 24X$

Assume, for contradiction inside this subsection, that

$$X \leq n_{\min} \leq 24X.$$

Rotate the cycle so that $n_1 = n_{\min}$, and write

$$n_1 = a_1 2^{k_1} - 1.$$

The finite certificate uses the five-step prefix

$$n_1 \rightarrow n_2 \rightarrow n_3 \rightarrow n_4 \rightarrow n_5 \rightarrow n_6, \quad d = 5.$$

Certified ranges for n_i and k_i . Write the certified size windows as

$$N_i^- \leq n_i \leq N_i^+.$$

For $m = 94$, the exact windows used by the certificate are:

$$i = 1. \quad N_1^- = X$$

$$N_1^+ = 24X$$

$$1 \leq k_1 \leq 75.$$

$$i = 2. \quad N_2^- = X$$

$$N_2^+ = 912400181570036563679542089616101765$$

$$1 \leq k_2 \leq 119.$$

$$i = 3. \quad N_3^- = X$$

$$N_3^+ = 822328213130442028404082204488730731843972603553922306122$$

$$1 \leq k_3 \leq 189.$$

$$i = 4. \quad N_4^- = 57049502556188971503921$$

$$N_4^+ = 1571423169375338536952767633230783186188564282618035280474448132226771450375422013698540295$$

$$1 \leq k_4 \leq 299.$$

$$i = 5. \quad N_5^- = 1168976708683560667551421091061578373$$

$$N_5^+ = 70401126729176769962360662850199050441155082223462196271358635493053593926837126274040422449098627806501217496449237262513197089319826978620616$$

$$1 \leq k_5 \leq 474.$$

$$i = 6. \quad N_6^- = 1465952061084188827016401157753797532004626278937197464086$$

$$N_6^+ = 20646022781602000906682855510685997468263370248430441106632478286519831381104874074513251180194545735304129565647188961327296848339349580189446762931473736513065712684841607601302354231852008920616965026939419206250170631492651$$

terminal layer.

Dependent ℓ_i -ranges, a_1 -intervals, and residue classes. Once the finite n_i, k_i -windows are fixed, the rest of the certificate is exactly the same as in the two preceding cases. For every fixed exponent prefix, the dependent ℓ_i -range is computed from (13); the intervals for a_1 obtained from (8) and (9) are intersected; and the exact residue classes given by (16) and (17) are intersected. Therefore any genuine branch in the assumed range would leave a nonempty terminal condition

$$[L, U] \cap \{a_1 : a_1 \equiv r \pmod{2^b}\}.$$

The enumeration covers all $k_1 = 1, \dots, 75$, together with all dependent later exponent ranges. After the interval and residue conditions are intersected, the depth-five candidate set is empty:

$$\mathcal{S}_5 = \emptyset.$$

The assumed range $X \leq n_{\min} \leq 24X$ is impossible, and hence

$$n_{\min} > 24X.$$

9.2. Final Contradiction from $n_{\min} > 24X$

Since $n_{\min} > 24X$, every local minimum satisfies $n_i > 24X$. Hence Lemma 3 gives

$$T(n_i) < \frac{3}{n_i} < \frac{3}{24X} = \frac{1}{8X}.$$

Choose a suffix-balanced block of length 89. A 94-cycle has 94 local minima, so there are $94 - 89 = 5$ local minima outside the block. Their combined contribution is less than

$$\frac{5}{8X}.$$

For a term inside the block, let t be the suffix length beginning at that term. Then $1 \leq t \leq 89$, and Lemma 5 gives

$$n_j + 1 \geq 2^{V_t}, \quad V_t = \frac{tK}{94} \frac{\delta - 1}{\delta^t - 1}.$$

Using $K \geq K_0$ and Lemma 9,

$$V_t \geq \frac{89K_0}{94} \frac{\delta - 1}{\delta^{89} - 1}.$$

An exact rational-arithmetic check shows that the floor of this last exponent is at least 179. Thus each of the 89 block terms satisfies

$$n_j \geq 2^{179} - 1, \quad T(n_j) < \frac{3}{2^{179} - 1}.$$

The total block contribution is therefore less than

$$\frac{89 \cdot 3}{2^{179} - 1} = \frac{267}{2^{179} - 1}.$$

Combining the omitted part and the block part gives

$$\sum_{i=1}^{94} T(n_i) < E_{94}, \quad E_{94} = \frac{5}{8X} + \frac{267}{2^{179} - 1}.$$

By Lemma 2,

$$\delta < \frac{K + L}{K} < \delta + \frac{E_{94}}{3K \log 2} \leq \delta + \frac{E_{94}}{3K_0 \log 2}.$$

An exact rational-arithmetic check gives

$$\frac{E_{94}}{3K_0 \log 2} < 6.191 \cdot 10^{-43}.$$

Hence

$$\delta < \frac{K + L}{K} < \delta + 6.191 \cdot 10^{-43}.$$

The continued-fraction denominator calculation shows that this interval contains no rational number whose positive reduced denominator is smaller than

$$Q = 7941964418702608664581.$$

Since the reduced denominator of $(K + L)/K$ divides K , the denominator test forces $K \geq Q$. But the Simons–de Weger upper bound and an exact rational-arithmetic check give

$$K < 1.4784 \cdot 94 \cdot \delta^{94} < Q.$$

This contradiction excludes $m = 94$.

10. The Case $m = 95$

By Lemma 12, every hypothetical 95-cycle satisfies

$$K \geq K_0 = 205632218873398596256.$$

We now combine this lower bound with the completed depth-six finite certificate. The proof again has two parts. First, the finite certificate rules out

$$X \leq n_{\min} \leq 24X.$$

Then the improved bound $n_{\min} > 24X$ is inserted into the product inequality to obtain the final continued-fraction contradiction.

10.1. Exclusion of the Range $X \leq n_{\min} \leq 24X$

Assume, for contradiction inside this subsection, that

$$X \leq n_{\min} \leq 24X.$$

Rotate the cycle so that

$$n_1 = n_{\min}, \quad n_1 = a_1 2^{k_1} - 1.$$

The finite certificate uses the six-step prefix

$$n_1 \rightarrow n_2 \rightarrow n_3 \rightarrow n_4 \rightarrow n_5 \rightarrow n_6 \rightarrow n_7, \quad d = 6.$$

Certified ranges for n_i and k_i . Write the certified size windows as

$$N_i^- \leq n_i \leq N_i^+.$$

For $m = 95$, the exact windows used by the certificate are:

$$i = 1. \quad N_1^- = X$$

$$N_1^+ = 24X$$

$$1 \leq k_1 \leq 75.$$

$$i = 2. \quad N_2^- = X$$

$$N_2^+ = 912400181570036563679542089616101765$$

$$1 \leq k_2 \leq 119.$$

$$i = 3. \quad N_3^- = X$$

$$N_3^+ = 822328213130442028404082204488730731843972603553922306122$$

$$1 \leq k_3 \leq 189.$$

$$i = 4. \quad N_4^- = X$$

$$N_4^+ = 1571423169375338536952767633230783186188564282618035280474448132226771450375422013698540295$$

$$1 \leq k_4 \leq 299.$$

$$i = 5. \quad N_5^- = 57049502556188967157335$$

$$N_5^+ = 70401126729176769962360662850199050441155082223462196271358635493053593926837126274040422449098627806501217496449237262513197089319826978620616$$

$$1 \leq k_5 \leq 474.$$

$$i = 6. \quad N_6^- = 1168976708683560443767721128513922951$$

$$N_6^+ = 20646022781602000906682855510685997468263370248430441106632478286519831381104874074513251180194545735304129565647188961327296848339349580189446762931473736513065712684841607601302354231852008920616965026939419206250170631492651$$

$$1 \leq k_6 \leq 751.$$

$i = 7$. $N_7^- = 1465952061084188122286096719636172011047425890134862921439$
 $N_7^+ = 3625533756011800758057383042766290425163692075458009640853985667833356424508$
 $172443956658603509921266241835468987222617944027516962708816880701465572241657794$
 $691209573997745278488466138052083272083918730947010371463518990101966566975031825$
 $1919016959114145169444768862115207853136293229952227096436755773806618681607973581$
 $959111109037578762200652342648606232340$
 terminal layer.

Dependent ℓ_i -ranges, a_1 -intervals, and residue classes. After each k_i -choice, the range of ℓ_i is computed from (13). For every complete prefix, all size constraints are pulled back to intervals for the same initial integer a_1 by (8) and (9). The two exact divisibility tests are then merged as the residue conditions (16) and (17). Thus a surviving branch would again give one nonempty intersection

$$[L, U] \cap \{a_1 : a_1 \equiv r \pmod{2^b}\}.$$

The full-range enumeration covers all $k_1 = 1, \dots, 75$, together with all dependent later exponent ranges. After the interval and residue conditions are intersected, the depth-six candidate set is empty:

$$\mathcal{S}_6 = \emptyset.$$

This contradicts the assumed range $X \leq n_{\min} \leq 24X$. Hence

$$n_{\min} > 24X.$$

10.2. Final Contradiction from $n_{\min} > 24X$

Since $n_{\min}$ is the least local minimum, the preceding inequality gives

$$n_i > 24X \quad (1 \leq i \leq 95).$$

Choose a suffix-balanced block of length 90. The five local minima outside this block contribute, by Lemma 3, less than

$$\frac{15}{24X}$$

to $\sum_i T(n_i)$. For the selected block, suffix-balance, Lemma 5, and Lemma 9 give, for every term in the block,

$$n_j + 1 \geq 2^{(90K/95)(\delta-1)/(\delta^{90}-1)} \geq 2^{(90K_0/95)(\delta-1)/(\delta^{90}-1)}.$$

The exact rational calculation gives

$$\frac{90K_0}{95} \frac{\delta-1}{\delta^{90}-1} > 113.$$

Hence each of the 90 selected terms contributes less than

$$\frac{3}{2^{113} - 1},$$

and therefore

$$\sum_{i=1}^{95} T(n_i) < E_{95},$$

where

$$E_{95} = \frac{15}{24X} + \frac{270}{2^{113} - 1}.$$

Substituting this into Lemma 2, and using $K \geq K_0$, gives

$$\delta < \frac{K + L}{K} < \delta + \frac{E_{95}}{3K_0 \log 2}.$$

The exact rational calculation gives

$$\frac{E_{95}}{3K_0 \log 2} < 6.191 \cdot 10^{-43}.$$

This is smaller than the certified continued-fraction width $6.2442 \cdot 10^{-43}$, so Lemma 10 gives

$$K \geq Q, \quad Q = 7941964418702608664581.$$

On the other hand, the Simons–de Weger upper bound gives

$$K < 1.4784 \cdot 95 \cdot \delta^{95} < Q,$$

again by exact rational arithmetic. This contradiction excludes $m = 95$.

11. Main Theorem

Theorem 14. *Assuming the certified inputs in Section 2, there is no non-trivial Collatz m -cycle with $m \leq 95$.*

Proof. The cases $m \leq 91$ are excluded by Lemma 13. The case $m = 92$ is excluded in Section 7. The case $m = 93$ is excluded in Section 8. The case $m = 94$ is excluded in Section 9. The case $m = 95$ is excluded in Section 10. These alternatives cover all positive integers $m \leq 95$. Therefore no non-trivial Collatz m -cycle exists with $m \leq 95$. $\square$

Remark 15. The proof above does not use Hercher’s consecutive-block contribution estimate. Instead, the suffix-balanced block selection gives lower bounds for the terms of each chosen block directly from suffix odd-step sums. This replacement is used in all applications of the block estimate.

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